

Iván Cárdenas

Geospatial Specialist · PhD Candidate

[linkedin.com/in/ivancardenasleon](https://www.linkedin.com/in/ivancardenasleon)

ilcardenas26@gmail.com

+31 0686185983

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Planet Labs PBC

Haarlem, Netherlands

Job Application for Solutions Architect

Dear Planet Hiring Team,

About Me:

When I was young, I spent hours just looking at the world atlas we had at home. Even though I knew most of the pages, I kept staring at the terrain and thematic maps. Though I did not go straight to cartography or Earth observation, those early hours shaped my curiosity. My background is a **BEng. in Environmental and Sanitary**, I went on to study a **PG-Diploma in GIS** at Colombia's National Geographic Institute and later completed a **Master's degree in Geo-information Science** at the University of Twente, Netherlands. I am now in the third year of my **PhD in Urban Digital Twins**, also at Twente, where I work on translating complex, multi-source spatial data into decisions that actually reach planners and policymakers.

In my free time, I play tycoon and strategy videogames, play beach volleyball, go hiking, find creative inspiration from illustrators or take photos of nature.

My experience in Geospatial & Earth Observation:

Over eight years of applied geospatial work I have moved from hydrological modelling (HEC-HMS, HEC-RAS) to building full-stack spatial data platforms. In my current research I am developing **CrossTwin**, an open-source Urban Digital Twin platform built on GeoDjango, PostGIS, React, and TITiler, with a DAG-based spatial query architecture across three urban sectors: drinking water supply, heat stress, and housing provision. Alongside this, I have built and deployed real operational tools — a Streamlit web application for Circular Energy scenario analysis, a Python Panel dashboard for Vitens, the Dutch drinking water authority, and a ArcGIS Experience and geoprocessing tool for heat stress — which allows for scenario-based modelling, impact assessment, and environmental costs. These tools were designed for planners and engineers who are not software developers, which taught me early that the most technically elegant solution fails if it does not speak to the stakeholder's actual question.

Beyond research, I have delivered applied geospatial solutions in consulting roles across Colombia and the Netherlands, including UAV imagery pipelines, environmental impact assessments, and Web-GIS development using Mapbox GL and JavaScript. At Philips, I replaced a manual biodiversity spreadsheet workflow with an interactive global manufacturing site map integrating satellite and API-based land-use data, a project that, I believe, closely mirrors the kind of customer-facing solution design Planet's Professional Services team produces every day.

I am proficient in Python for geospatial applications (GeoPandas, rasterio, GDAL), experienced with QGIS and ArcGIS, and comfortable working in Agile cross-functional teams. I regularly use AI tools including Claude code and other LLM-assisted coding to accelerate prototyping and demo development. I have also designed and delivered university courses on Digital Twins and Urban Heat Modelling, and soon to be a speaker at ISPRS international conferences, skills that reflect both my technical depth and my ability to communicate complex methods to diverse audiences.

Why I want to join Planet:

Planet's mission of making global change visible and actionable resonates directly with the core argument of my research: that geospatial data only generates societal value when it is connected to a decision. Working with a direct provider of satellite imagery and other geospatial data, would be an ideal opportunity to put my skills to use in a way that is both useful and impactful. I would welcome the opportunity to bring that perspective to your customers, your pre-sales process, and your Professional Services team in Haarlem.

Thank you for your time and consideration. I look forward to the possibility of contributing to Planet's work.

Sincerely,

Iván Cárdenas-León

PhD Candidate in Urban Planning, University of Twente

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